

# MVP Specifications

# Altrea Hackathon MVP: Development Roadmap

## 1. System Architecture Overview

- **Role Switcher:** A landing screen or side menu to toggle between **Caregiver View** and **Elder View**.
- **Data Source Simulation:** Since we are simulating stress, the app needs a "Demo Mode" toggle that pushes fake "high-stress" EEG data to trigger the alert system.
- **Input Types:** The app processes two data streams:
  1. **Passive:** Simulated EEG waves (Alpha, Beta, Theta).
  2. **Active:** Manual "Self-Select" mood input from the Elder.
- **Colors:** White, light colors, light pink, light purple, light blue (Calm)

## 2. User Interface: Caregiver View (The Monitor)

*The goal is "Peace of Mind" through data visibility.*

- **Live EEG Stream:**
  - Real-time moving line graph showing brainwave activity.
  - Visual indicators for Alpha (relaxation), Beta (focus/stress), and Theta (drowsiness/meditation).
- **Historical Dashboard:**
  - Daily/Weekly view of "Emotional Thresholds" (e.g., % of time spent in 'Lonely' vs. 'Calm').
- **Alert & Notification Center:**
  - **Past Alerts:** Log of time-stamped stress events.
  - **Active Insights:** A pop-up notification when stress is detected: *"Warning: High Stress detected. Insight: Cognitive fatigue. Suggested Intervention: Trigger 'Guided Breathing' or call now."*
- **Circle of Care (Contact Page):**
  - Centralized directory with "One-Tap Call" buttons for: Primary Caretaker, Family Members, Doctors, and Emergency Services.

## 3. User Interface: Elder View (The Companion)

*The goal is simplicity, high contrast, and ease of use.*

- **Self-Select Mood (Daily Check-in):**

- **Step 1:** "How are you feeling today?" (Two big buttons: **GOOD** vs. **BAD**).
- **Step 2 (If Bad):** Options for **Stressed, Lonely, or Sad** (Dark Blue/Dark Red/Dark Purple UI).
- **Step 2 (If Good):** Options for **Happy or Calm** (Green/Yellow UI).
- **AI Intervention Engine:**
  - After selection, display a suggested activity:
    - *Play a Song* (Music Player UI).
    - *Meditation* (Audio guide).
    - *Call Family* (Quick dial).
    - *Memory/Gratitude Sharing* (Prompt to record a voice note or view a photo).
- **Personal Stats:**
  - Simplified Live EEG view (so they feel connected to their own health).
  - Simple history (e.g., "You've had 4 calm days this week!").

## 4. Key Demo Functions

### Scenario A: The Stress Alert (Passive Detection)

1. **Action:** In the dev backend, trigger "Simulation: High Stress."
2. **Visual:** In the **Caregiver View**, the EEG waves move from smooth (Alpha) to erratic/high frequency (Beta).
3. **Result:** A notification pops up on the Caregiver's phone: *"High stress detected in [Name]. Suggesting 'Calm Playlist' intervention."*

### Scenario B: Manual Support (Active Detection)

1. **Action:** The Elder clicks "I'm feeling lonely."
2. **Visual:** UI changes to Dark Blue. The app suggests: *"Would you like to hear a gratitude message from your granddaughter?"*
3. **Result:** The Elder plays the message, and the Caregiver receives a log update: *"Elder initiated 'Memory Sharing' due to loneliness."*

| Emotions (Input) | Interventions (Output)                                                                                                            |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Happy            | Capture this moment - take a picture.<br>Gratitude journal                                                                        |
| Calm             | Play nature sounds                                                                                                                |
| Stressed         | Breathing exercise "Follow the circle: Breathe in for 4s, hold for 4s, out for 4s" (It would be cool if the app can display this) |

|        |                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------|
|        | Automatically play a “Zen” music list.                                                           |
| Lonely | Show a video message previously recorded by a loved one.<br>Call a family member or notify them. |
| Sad    | Show memories - pictures or videos                                                               |

## **Feedback:**

### **1. Two distinct views + Account/Role Switcher (required)**

- a. Altea has two separate user experiences:
  - i. Caregiver View (Monitor / Peace of Mind)
  - ii. Elder View (Companion / Simplicity & Support)
- b. We need an explicit Account/Role Switcher (landing screen or side menu) to toggle between:
  - i. Caregiver account
  - ii. Elder account

(This is essential for the demo and for validating both user journeys)

### **2. Caregiver View**

- a. In the link shared we already have:
  - i. EEG Live: Real-time *moving* line graph + indicators for **Alpha/Beta/Theta**.
  - ii. Historical data: Daily/Weekly breakdown of emotional thresholds (e.g., % Calm vs Lonely).
  - iii. Alerts: Past alerts and notifications.
- b. “Caregivers & Family”:
  - i. We don't want to include the word “Emergency”, just the button to call: “One-Tap Call” buttons for: Primary Caretaker, Family Members, Doctors, and Emergency Services.

### **3. Elder View**

- a. Elder Login / Home Flow (required)

When the elder logs in, the flow should be:

1. Question first (Self-Select Mood):  
“How are you feeling today?”
  - Two big buttons: GOOD / BAD
  - If BAD → Stressed / Lonely / Sad
  - If GOOD → Happy / Calm
2. Then read EEG (Passive input):  
Show simplified EEG view (Alpha/Beta/Theta) so the elder feels connected.
3. Then provide recommendations (Interventions):  
Based on the combination of:
  - Elder’s selected mood (Active input)
  - EEG state (Passive input)

# Notes

Important points to consider in the story

1. Why we put in these features -> Not just saying why and how this is beneficial to the customers but saying that we thought about various factors as we designed the product.  
-> Throw words like pedagogical

### Managing cognitive load is key to understanding, retention and learning

|               |
|---------------|
| ✓ Simple      |
| ✓ Chunked     |
| ✓ Analogies   |
| ✓ Visual aids |
| ✓ Real-world  |

|                |
|----------------|
| ✗ Repetition   |
| ✗ Redundancy   |
| ✗ Confusing    |
| ✗ Overwhelming |
| ✗ Wall of text |

### Curiosity is a powerful driver of learning

✓ Encouraging tone

✓ Challenges

✓ Relevance

✓ Autonomy

✗ Neutral tone

✗ Too easy or difficult

✗ No meaning

✗ No choices

2. We are also considerate of Sustainable AI. So instead of LLMs we use small processing models (can dig deeper into this)
3. We are wearable agnostic. We want to be able to sync to all wearables. Many wearable just give out the signal. We interpret the signal, process it, support to overcome/cherish the emotion and also provide the signal to the caregivers
- 4.

### Patient View vs Caretaker View

- Caretaker view
  - Live EEG view
  - Dashboard / Insights - Weekly overview
  - Notifications / Alerts for Caretaker
- Elder view - not too complicated, text is bigger
  - Main view is "How are you feeling today" and the activities (currently on the Insights page)
  - Centralized contact information page - caretaker, family / friends, doctor providers
  - (1) Passive EEG Data: Suggest interventions according to the data
  - (2) Active Self-select an option: I'm feeling sad. Let's see what this app will suggest.
    - Is today a good day or a bad day?
    - If it's a good day (green or yellow) -> Happy, Calm
    - If it's a bad day (blue or red) -> Stressed, Lonely, Sad
  - Second characteristic: How do I feel?
  - **Interventions** - should be for the elder
- Other activities: Should play this song, do a meditation exercise, calling, voice memo
  - Song

- Meditation
- Call
- Voice Memo
- Memory Sharing
- Gratitude Sharing / Journal

**Use Cases**

- Connecting caretaker and patient to each other
- Tracking for doctor
- Building a history
- Listening to music or song - that they start listening while they receive a song